

Independent Event

$$P(A \cap B) = P(A) \cdot P(B)$$

$$\text{on, } P(A|B) = P(A), P(B|A) = P(B)$$

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{if } P(B) \neq 0$$

Independence

A & B are independent if,

$$P(A|B) = P(A) \text{ or } P(B|A) = P(B)$$

$$\text{or } \boxed{P(A \cap B) = P(A) P(B)}$$

Conditional Independence

{ if three variables are given. }

A and B are conditionally independent if given C,

$$P(A, B|C) = P(A|C) \cdot P(B|C) \dots (i)$$

$$\Rightarrow P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$\text{Conditioning on C, } P(A|B, C) = \frac{P(A \cap B|C)}{P(B|C)} = \frac{P(A|C) P(B|C)}{P(B|C)}$$

$$\boxed{P(A, B|C) = P(A|C) \cdot P(B|C)}$$

Chain Rule

$$P(\underbrace{\text{Toothache}}_A, \underbrace{\text{Catch, Cavity}}_B)$$

'Conditional Independence related questions' ↗

↗ Normalization

'Compute distribution on query variable by fixing evidence variables & summing over hidden variables'

	toothache		¬toothache	
	catch	¬catch	catch	¬catch
cavity	.108	.012	.072	.008
¬cavity	.016	.064	.144	.576

Joint probability distribution sum would be '1'

if $> 1 \rightarrow$ Normalize

if $\leq 1 \rightarrow$ ¬ Normalize

Normalization denominator is ' α ' (Alpha)

most imp(1) Bayes' Rule ← exam 2
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$$P(A|B) = \frac{P(A \cap B)}{P(B)} \Rightarrow P(A \cap B) = P(A|B) P(B)$$

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

$$\Rightarrow \boxed{P(B|A) = \frac{P(A|B) P(B)}{P(A)}}$$

Bayes' Rule is based on conditional probability

Meningitis

$$\rightarrow P(S|M) = 0.5$$

→ causes stiff neck = 50%, ~~$P(S) = 0.05$~~

Prob of having meningitis = $P(M) = 1/50000$

Prob of stiff neck = $P(S) = 1/20$

Q: What's the probability of meningitis given stiff neck → $P(M|S)$

$$P(M|S) = \frac{P(S|M) P(M)}{P(S)} = \frac{0.5 * (1/50000)}{1/20}$$

$$= 0.0002$$

Shayan

Example

$$P(S|M) = 0.5$$

$$P(\cancel{W|M}) = 0.8$$
$$S|W$$

$$P(M|S) = \frac{P(S|M) P(M)}{P(S)}$$

$$P(W|S) = \frac{P(S|W) P(W)}{P(S)}$$

Given $P(S|W)$ & $P(W) = 1/1000$

$$\frac{P(M|S)}{P(W|S)} = \frac{P(S|M) P(M)}{P(S|W) P(W)} = \frac{.5 \times \frac{1}{50000}}{.8 \times \frac{1}{1000}} = \frac{1}{80}$$

↗ Bayes' Rule &
Conditional Independence

Pneumonia causes fever, $P(F|P) = 0.95$

Probability of person having Pneumonia, $P(P) = 1 \times 10^{-7}$

What is the probability that pneumonia is the cause of fever?

$$\rightarrow P(P|F)$$

$$P(F) = 1/100$$

$$P(P|F) = \frac{P(F|P) P(P)}{P(F)}$$

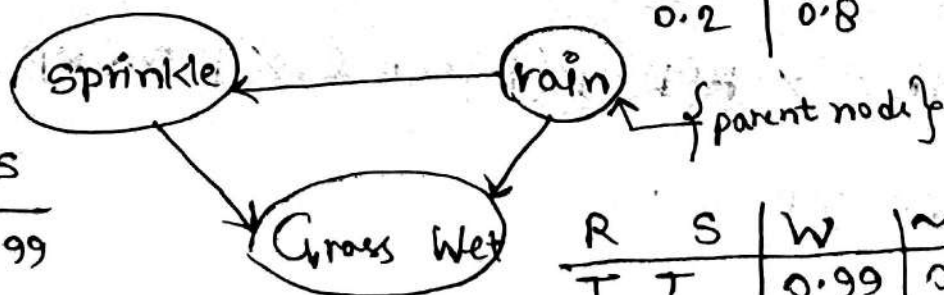
$$= \frac{0.95 \times 1 \times 10^{-7}}{1/100}$$

$$= 9.5 \times 10^{-6}$$

Reasoning Under Uncertainty

Bayesian Network

- Probabilistic graph model helps to deal with uncertainty.
- Each node: Random variable [DAG] = Directed Acyclic Graph
- Arc: Dependency



R	~R
0.2	0.8

{parent node}

R	S	~S
T	0.01	0.99
F	0.4	0.6

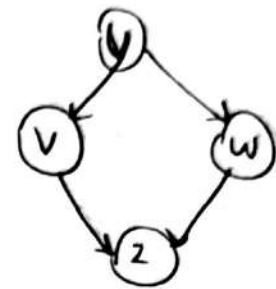
R	S	W	~W
T	T	0.99	0.01
T	F	0.9	0.1
F	T	0.8	0.2
F	F	0.0	1.00

Q: Find $P(W, S, R)$

$$= P(W|S, R) \times P(S|R) \times P(R)$$

$$= \cancel{0.01} \times 0.99 \times 0.01 \times 0.2$$

$$= 1.98 \times 10^{-3}$$



$$\Rightarrow P(U, V, W, Z)$$

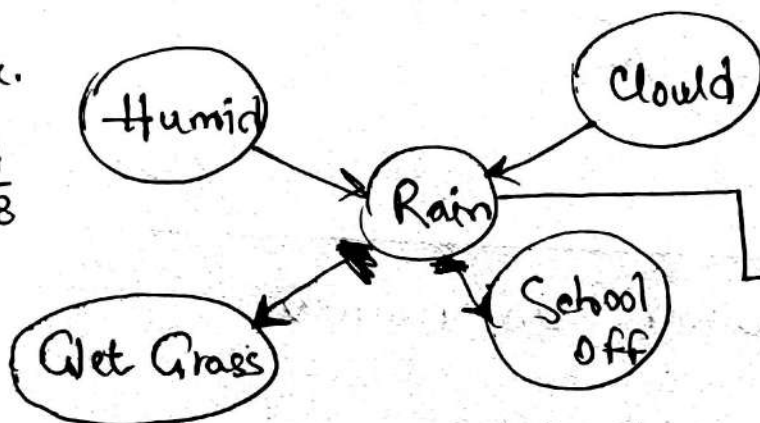
Q: Find $P(W, \neg S, \neg R)$

$$P(W|\neg S, \neg R) \times P(\neg S|\neg R) \times P(\neg R)$$

$$= 0.00$$

Ex.

H	$\neg H$
0.2	0.8



C	$\neg C$
0.6	0.4

H	C	R	$\neg R$
T	T	0.9	0.1
T	F	0.4	0.6
F	T	0.7	0.3
F	F	0.1	0.9

R	W	$\neg W$
T	0.8	0.2
F	0.1	0.9

R	0	$\neg 0$
T	0.7	0.3
F	0.2	0.8

Q: Find $P(0, R, \neg H, C)$

$$= P(0|R, \neg H, C) \times P(R|\neg H, C) \times P(\neg H) \times P(C)$$

$$= 0.7 \times 0.7 \times 0.8 \times 0.6$$

$$= 0.2352$$